Unit: Unit 3: Functions
Topic: Inverse Relations and Functions

Name: ____________
Date: ________

Homework (Carolina Reaper)

Q1. A music composer uses a function $f(x) = 3x + 2$ to create rhythm patterns. Which of the following is the inverse function?
(A) $f^{-1}(x) = \frac{x-2}{3}$
(B) $f^{-1}(x) = \frac{x+2}{3}$
(C) $f^{-1}(x) = 3x - 2$
(D) $f^{-1}(x) = \frac{x}{3} - 2$
(E) $f^{-1}(x) = \frac{2-x}{3}$

Q2. The frequency of a sound wave is given by $f(x) = 2x^2 + 1$. Does this function have an inverse?
(A) Yes, it has a unique inverse
(B) No, it does not have an inverse
(C) Maybe, it depends on the domain
(D) Only for positive values of x
(E) Only for negative values of x

Q3. An art canvas has a width given by $f(x) = x^2 - 4$. Find the inverse function.
(A) $f^{-1}(x) = \sqrt{x+4}$
(B) $f^{-1}(x) = \sqrt{x-4}$
(C) $f^{-1}(x) = x^2 + 4$
(D) $f^{-1}(x) = \pm\sqrt{x+4}$
(E) $f^{-1}(x) = \pm\sqrt{x-4}$

Q4. The duration of a music piece is given by $f(x) = \frac{1}{x}$. Verify if $f^{-1}(x) = \frac{1}{x}$ is the inverse function using composition.
(A) Yes, it is the inverse
(B) No, it is not the inverse
(C) Maybe, it depends on the domain
(D) Only for positive values of x
(E) Only for negative values of x

Q5. A painter uses a function $f(x) = x^3 - 1$ to create unique patterns. Find the inverse function.
(A) $f^{-1}(x) = \sqrt[3]{x+1}$
(B) $f^{-1}(x) = \sqrt[3]{x-1}$
(C) $f^{-1}(x) = x^3 + 1$
(D) $f^{-1}(x) = x^2 - 1$
(E) $f^{-1}(x) = \frac{1}{x^3} + 1$

Q6. A musician uses a function $f(x) = \frac{1}{x-1}$ to create sound effects. Does this function have an inverse?
(A) Yes, it has a unique inverse
(B) No, it does not have an inverse
(C) Maybe, it depends on the domain
(D) Only for positive values of x
(E) Only for negative values of x

Q7. An artist uses a function $f(x) = 2x - 5$ to create unique designs. Find the inverse function.
(A) $f^{-1}(x) = \frac{x+5}{2}$
(B) $f^{-1}(x) = \frac{x-5}{2}$
(C) $f^{-1}(x) = x + 5$
(D) $f^{-1}(x) = 2x + 5$
(E) $f^{-1}(x) = \frac{x}{2} - 5$

Q8. A sound engineer uses a function $f(x) = x^2$ to adjust sound levels. Does this function have an inverse?
(A) Yes, it has a unique inverse
(B) No, it does not have an inverse
(C) Maybe, it depends on the domain
(D) Only for positive values of x
(E) Only for negative values of x

Open Ended Questions (FRQ)

Q9. A music composer uses a function $f(x) = 2x^2 + 3x - 1$ to create rhythm patterns. Find the inverse function and verify it using composition. Show all your work.

Q10. An artist uses a function $f(x) = x^3 - 2x^2 - 5x + 1$ to create unique designs. Determine if this function has an inverse and explain your reasoning. Show all your work.

Chef's Tips

- Tip 1: Remind students to check their work carefully
- Tip 2: Encourage students to use graphing calculators to visualize the functions
- Tip 3: Emphasize the importance of showing all work and explanations for free response questions